



Semester : 19991

Faculty : College of Computer & Information Sc

Plan Hrs. : 134

Degree : Bachelor

Department : COMPUTER SCIENCE

Student Level : Earned Hrs.

Plan Type : Major

Major : CS-DX COMPUTER SCIENCE (D

Gender : All

Study Type Hours

Edition : 1

Category : Univ\_Req 23 Hrs

Compulsory (1) 23 Hrs

| First   |                                     |     |           |
|---------|-------------------------------------|-----|-----------|
| Code    | Course Name                         | Crd | requisite |
| ARAB101 | Arabic Writing                      | 2   |           |
| ENG101  | Intensive Writing                   | 3   |           |
| ISC101  | Islamic Ethics                      | 2   |           |
| Second  |                                     |     |           |
| Code    | Course Name                         | Crd | requisite |
| ISC103  | Islamic Economic System             | 2   |           |
| Third   |                                     |     |           |
| Code    | Course Name                         | Crd | requisite |
| COM201  | Communication Skills                | 3   |           |
| ISC105  | Studies of the Holy Qu'ran          | 2   |           |
| Fourth  |                                     |     |           |
| Code    | Course Name                         | Crd | requisite |
| ARAB103 | Professional and Literary Writing   | 2   |           |
| PSY101  | Introduction to Psychology          | 3   |           |
| Sixth   |                                     |     |           |
| Code    | Course Name                         | Crd | requisite |
| ISC203  | Contemporary Financial Transactions | 2   |           |
| Seventh |                                     |     |           |
| Code    | Course Name                         | Crd | requisite |
| ARAB203 | Arabic Grammar                      | 2   |           |

Category : College\_Req 46 Hrs

Compulsory (1) 44 Hrs

| First   |                                                   |     |               |
|---------|---------------------------------------------------|-----|---------------|
| Code    | Course Name                                       | Crd | requisite     |
| CS101   | Computer Programming-I                            | 4   |               |
| MATH111 | Calculus I                                        | 3   | MATH002 -     |
| Second  |                                                   |     |               |
| Code    | Course Name                                       | Crd | requisite     |
| CS102   | Computer Programming-II                           | 3   | CS101 - pre   |
| CS175   | Digital Logic and Computer Organization           | 3   |               |
| ENG103  | Research Writing Techniques                       | 3   | ENG101 - pre  |
| STAT101 | Introduction to Statistics and Probability Theory | 3   | MATH111 - pre |

| Third   |                                           |     |                            |
|---------|-------------------------------------------|-----|----------------------------|
| Code    | Course Name                               | Crd | requisite                  |
| CS210   | Data Structures and Algorithms            | 3   | CS102 - pre                |
| CS285   | Discrete Mathematics for Computing        | 3   | CS101 - pre                |
| Fourth  |                                           |     |                            |
| Code    | Course Name                               | Crd | requisite                  |
| CS330   | Introduction to Operating Systems         | 3   | CS175 - pre<br>CS210 - pre |
| Fifth   |                                           |     |                            |
| Code    | Course Name                               | Crd | requisite                  |
| CS331   | Data Communications and Computer Networks | 3   | CS175 - pre<br>CS210 - pre |
| ETHC303 | Ethical and Social Aspects of Computing   | 3   |                            |
| Eighth  |                                           |     |                            |
| Code    | Course Name                               | Crd | requisite                  |
| CS492   | Cooperative Training                      | 10  |                            |

Elective (1) 2 Hrs

| Code   | Course Name                   | Crd | requisite |
|--------|-------------------------------|-----|-----------|
| HPE101 | Health & Physical Education I | 1   |           |
| PE101  | Beginner Badminton            | 1   |           |
| PE103  | Beginner Basketball           | 1   |           |
| PE105  | Beginner Gymnastics           | 1   |           |
| PE107  | Beginner Handball             | 1   |           |
| PE109  | Beginner Soccer               | 1   |           |
| PE111  | Beginner Squash               | 1   |           |
| PE113  | Beginner Swimming             | 1   |           |
| PE115  | Beginner Table Tennis         | 1   | PED002 -  |
| PE117  | Beginner Tennis               | 1   |           |
| PE119  | Beginner Volleyball           | 1   |           |
| PE121  | Beginner Weight Training      | 1   |           |

| Code   | Course Name                    | Crd | requisite |
|--------|--------------------------------|-----|-----------|
| HPE102 | Health & Physical Education II | 1   |           |

| Code  | Course Name            | Crd | requisite |
|-------|------------------------|-----|-----------|
| PE201 | Intermediate Badminton | 1   |           |



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|       |                              |   |  |
|-------|------------------------------|---|--|
| PE203 | Intermediate Basketball      | 1 |  |
| PE205 | Intermediate Gymnastics      | 1 |  |
| PE207 | Intermediate Handball        | 1 |  |
| PE209 | Intermediate Soccer          | 1 |  |
| PE211 | Intermediate Squash          | 1 |  |
| PE213 | Intermediate Swimming        | 1 |  |
| PE215 | Intermediate Table Tennis    | 1 |  |
| PE217 | Intermediate Tennis          | 1 |  |
| PE219 | Intermediate Volleyball      | 1 |  |
| PE221 | Intermediate Weight Training | 1 |  |

|                                |          |            |
|--------------------------------|----------|------------|
| <b>Category :</b> CS_Electives | <b>3</b> | <b>Hrs</b> |
| <b>Elective (1 )</b>           | <b>3</b> | <b>Hrs</b> |

| Code  | Course Name                             | Crd | requisite |
|-------|-----------------------------------------|-----|-----------|
| SE423 | Software Engineering Project Management | 3   |           |

| Code   | Course Name                              | Crd | requisite                    |
|--------|------------------------------------------|-----|------------------------------|
| CS316  | Introduction to AI and Data Science      | 3   | CS210 - pre<br>STAT101 - pre |
| CS381  | System Programming                       | 3   | CS330 - pre                  |
| CS387  | Mobile Applications Development          | 3   | SE371 - pre                  |
| CS415  | Internet of Things (IoT)                 | 3   | CS331 -                      |
| CS439  | Search Engines and Information Retrieval | 3   | CS340 - pre                  |
| CS455  | Computational Bioinformatics             | 3   | CS311 - pre                  |
| CS460  | Introduction to Robotics                 | 3   | CS210 - pre                  |
| CS489  | Selected Topics in Computer Science      | 3   | CS210 -                      |
| CS495  | Emerging Topics in Computer Science      | 3   |                              |
| CS496  | Emerging Topics in AI and DS             | 3   |                              |
| DMS426 | Game Development                         | 3   |                              |
| IS361  | Project Management                       | 3   | IS231 - pre<br>IS241 -       |
| IS362  | Integrated Information                   | 3   | IS231 -                      |

|       |                                        |   |               |
|-------|----------------------------------------|---|---------------|
|       | Systems                                |   |               |
| IS371 | Quantitative Analysis                  | 3 | STAT101 - pre |
| IS465 | Geographic Information Systems         | 3 | BUS373 -      |
| IS487 | Emerging Topics in Information Systems | 3 |               |
| IS489 | Selected Topics in Information Systems | 3 |               |
| SE311 | Software Requirements Analysis         | 3 | SE201 - pre   |
| SE365 | Human Computer Interaction             | 3 | CS210 - pre   |
| SE409 | Cloud Computing                        | 3 | CS331 - pre   |
| SE480 | Software Analytics                     | 3 |               |
| SE489 | Special Topics in Software Engineering | 3 |               |

|                           |          |            |
|---------------------------|----------|------------|
| <b>Category :</b> Bus_Req | <b>3</b> | <b>Hrs</b> |
| <b>Elective (1 )</b>      | <b>3</b> | <b>Hrs</b> |

| Code    | Course Name                         | Crd | requisite |
|---------|-------------------------------------|-----|-----------|
| ACC111  | Foundations in Financial Accounting | 3   |           |
| BUS101  | Introduction to Business            | 3   |           |
| ECON101 | Principles of Microeconomics        | 3   |           |
| ECON102 | Principles of Macroeconomics        | 3   |           |

|                                   |           |            |
|-----------------------------------|-----------|------------|
| <b>Category :</b> CS Core Courses | <b>41</b> | <b>Hrs</b> |
| <b>Compulsory (1 )</b>            | <b>41</b> | <b>Hrs</b> |

| First   |                                  |     |               |
|---------|----------------------------------|-----|---------------|
| Code    | Course Name                      | Crd | requisite     |
| SCI101  | Introduction to Physical Science | 3   |               |
| Second  |                                  |     |               |
| Code    | Course Name                      | Crd | requisite     |
| MATH113 | Calculus II                      | 3   | MATH111 - pre |
| Third   |                                  |     |               |
| Code    | Course Name                      | Crd | requisite     |
| CS223   | Computational Linear             | 3   | MATH113 - pre |



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**Edition :** 1

|                |                                               |            |                             |
|----------------|-----------------------------------------------|------------|-----------------------------|
|                | Algebra                                       |            |                             |
| PHY105         | Physics I                                     | 4          |                             |
| <b>Fourth</b>  |                                               |            |                             |
| <b>Code</b>    | <b>Course Name</b>                            | <b>Crd</b> | <b>requisite</b>            |
| CS320          | Programming Languages: Concepts and Paradigms | 3          | CS210 - pre                 |
| PHY205         | Physics II                                    | 4          | PHY105 - pre                |
| SE201          | Introduction to Software Engineering          | 3          | CS102 - pre<br>ENG103 - pre |
| <b>Fifth</b>   |                                               |            |                             |
| <b>Code</b>    | <b>Course Name</b>                            | <b>Crd</b> | <b>requisite</b>            |
| CS311          | Design and Analysis of Algorithms             | 3          | CS210 - pre<br>CS285 - pre  |
| CS340          | Introduction to Database Systems              | 3          | CS210 - pre                 |
| <b>Sixth</b>   |                                               |            |                             |
| <b>Code</b>    | <b>Course Name</b>                            | <b>Crd</b> | <b>requisite</b>            |
| CS435          | Distributed Systems                           | 3          | CS330 - pre<br>CS331 - pre  |
| SE371          | Web Engineering                               | 3          | CS210 - pre                 |
| <b>Seventh</b> |                                               |            |                             |
| <b>Code</b>    | <b>Course Name</b>                            | <b>Crd</b> | <b>requisite</b>            |
| CS499          | Senior Project                                | 3          | CS498 -                     |
| CYS401         | Fundamentals of Cybersecurity                 | 3          |                             |

|                   |                         |          |            |
|-------------------|-------------------------|----------|------------|
| <b>Category :</b> | <b>Elective Courses</b> | <b>6</b> | <b>Hrs</b> |
|                   | <b>Elective (1 )</b>    | <b>6</b> | <b>Hrs</b> |

|             |                               |            |                  |
|-------------|-------------------------------|------------|------------------|
| <b>Code</b> | <b>Course Name</b>            | <b>Crd</b> | <b>requisite</b> |
| ACT101      | The Basics of Stage Acting    | 3          |                  |
| ART100      | Drawing                       | 2          |                  |
| CULT101     | Saudi Folklore and Culture I  | 3          |                  |
| CULT102     | Saudi Folklore and Culture II | 3          |                  |

|             |                    |            |                  |
|-------------|--------------------|------------|------------------|
| <b>Code</b> | <b>Course Name</b> | <b>Crd</b> | <b>requisite</b> |
| ART102      | Painting           | 2          |                  |

|         |                                      |   |              |
|---------|--------------------------------------|---|--------------|
| ART103  | Handcrafting                         | 2 |              |
| ART104  | Aesthetics of Arabic Calligraphy     | 2 |              |
| ART105  | Digital Media Design                 | 2 |              |
| ART106  | Introduction to Animation Techniques | 2 |              |
| ART107  | JEWELRY AND ACCESSORY DESIGN         | 2 |              |
| ASIA101 | Introduction to Japanese             | 3 |              |
| ASIA103 | Elementary Korean Language           | 3 |              |
| ASIA104 | Elementary Chinese Language          | 3 |              |
| CHM101  | General Chemistry                    | 4 |              |
| ELE103  | Stress Management                    | 3 |              |
| ELE109  | Public Relations                     | 3 |              |
| ELE110  | Entrepreneurship                     | 3 |              |
| ELP203  | The Language of Advertising          | 3 |              |
| ELP205  | Language and Digital Media           | 3 |              |
| EM101   | Engineering Drawing                  | 2 |              |
| ENG113  | Advanced Academic Reading            | 3 | ENG103 -     |
| ENG114  | Introduction to Journalism           | 3 | ENG103 - pre |
| EUR101  | European Language I                  | 3 |              |
| FRN101  | Elementary French                    | 3 |              |
| GER101  | Elementary German 1                  | 3 |              |
| HIST151 | History of Law                       | 3 |              |
| HIST153 | Comparative Political Systems        | 3 |              |
| IR101   | Information Resources                | 3 |              |
| LAW101  | Principles Of Law                    | 3 |              |
| LAW150  | Administrative Law I                 | 3 |              |
| LAW170  | Family Law                           | 2 |              |
| MATH101 | Finite Mathematics                   | 3 | MATH002 -    |

|             |                            |            |                  |
|-------------|----------------------------|------------|------------------|
| <b>Code</b> | <b>Course Name</b>         | <b>Crd</b> | <b>requisite</b> |
| AVN291      | Air Transport Geography    | 3          |                  |
| AVN292      | Aviation and Tourism       | 3          |                  |
| COM301      | Professional and Technical | 2          | COM201 -         |



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|        |                                    |   |          |
|--------|------------------------------------|---|----------|
|        | Presentations                      |   |          |
| ELP306 | English for Fashion Professionals  | 3 |          |
| ELP309 | Film Criticism                     | 3 |          |
| ENG207 | Critical Thinking                  | 3 | ENG103 - |
| ENG208 | Elementary Spanish                 | 3 |          |
| ENG209 | Intermediate Spanish               | 3 | ENG208 - |
| ENG301 | Technical Writing and Translation  | 3 | ENG101 - |
| ENG302 | English for Business Communication | 3 | ENG103 - |
| ISC205 | Family in Islam                    | 2 |          |

|  |               |
|--|---------------|
|  | STAT101 - pre |
|--|---------------|

|                              |               |
|------------------------------|---------------|
| <b>Category :</b> DX-Courses | <b>12 Hrs</b> |
|------------------------------|---------------|

|                        |              |
|------------------------|--------------|
| <b>Compulsory (1 )</b> | <b>6 Hrs</b> |
|------------------------|--------------|

| Fifth |                                     |     |           |
|-------|-------------------------------------|-----|-----------|
| Code  | Course Name                         | Crd | requisite |
| IS450 | Digital Transformation Fundamentals | 3   |           |

| Sixth |                                              |     |             |
|-------|----------------------------------------------|-----|-------------|
| Code  | Course Name                                  | Crd | requisite   |
| IS452 | Change Management for Digital Transformation | 3   | IS450 - pre |

|                      |              |
|----------------------|--------------|
| <b>Elective (1 )</b> | <b>6 Hrs</b> |
|----------------------|--------------|

| Code   | Course Name                                             | Crd | requisite   |
|--------|---------------------------------------------------------|-----|-------------|
| BCE483 | Business Intelligence and Analytics                     | 3   | IS450 - pre |
| IS453  | Technology Trends in Digital Transformation             | 3   | IS450 - pre |
| IS454  | Product Lifecycle Management for Digital Transformation | 3   | IS450 - pre |

| Code   | Course Name                                      | Crd | requisite    |
|--------|--------------------------------------------------|-----|--------------|
| CYS403 | Security Risk Management, Governance and Control | 3   | CYS401 - pre |
| IS446  | Knowledge Discovery and Data Mining              | 3   | IS450 - pre  |
| IS463  | Applied Data Science                             | 3   | CS210 - pre  |